

Discrete Structures (CS1005)

Course Instructor(s):

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Section(s): (All CS sections)

Final Exam

Total Time (Hrs): 3

Total Marks: [Click or tap here to enter text.](#)

Total Questions: 5

Date: May 23, 2026

Roll No

Course Section

Student Signature

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Attempt all the questions.
Solution

[CLO 1: Demonstrate the basic concepts of programming]

Q2:

[20 marks]

Read each of the following arguments carefully. Determine using the rules of inference whether each argument is **Valid** or **Invalid**. If the argument is invalid, provide a counterexample to justify your answer.

Argument	Premises (Symbolic Form)	Conclusion (Symbolic Form)	Valid / Invalid	Reason / Counterexample
1	1. $p \rightarrow q$ 2. $q \rightarrow r$ 3. $r \vee s$ 4. $\neg s$	$p \rightarrow \neg r$	Invalid	From $r \vee s$ and $\neg s$, we get r . Also, $p \rightarrow q$ and $q \rightarrow r$ imply $p \rightarrow r$, not $p \rightarrow \neg r$. Counterexample: $p=T, q=T, r=T, s=F$. Premises are true, conclusion is false.
2	1. $a \rightarrow (b \vee c)$ 2. $\neg b$ 3. $c \rightarrow d$	$a \rightarrow d$	Valid	Assume a . Then $b \vee c$. Since $\neg b$, by Disjunctive Syllogism we get c . From $c \rightarrow d$, conclude d . Therefore, $a \rightarrow d$.
3	1. $(m \wedge n) \rightarrow k$ 2. $\neg k \vee p$	$\neg n$	Invalid	Counterexample: $m=F, n=T, k=F$,

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	3. $p \rightarrow (m \vee n)$ 4. $\neg m$			$p=T$. All premises are true, but $\neg n$ is false. Therefore, the argument is invalid.
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Part b) Translate the following English statement into symbolic form and then simplify it using equivalent laws.

“It is not true that either Ali does not study hard and passes the exam, or Ali does not study hard and does not pass the exam; or Ali studies hard and passes the exam.”

Let:

- p : “Ali studies hard.”
- q : “Ali passes the exam.”

1. Write the statement in symbolic form.
2. Simplify the symbolic expression using equivalent laws.
3. State the final simplified expression.

Step	Expression	Law / Reason Used
1	$\neg[(\neg p \wedge q) \vee (\neg p \wedge \neg q) \vee (p \wedge q)]$	Symbolic form of the statement
2	$\neg[(\neg p \wedge (q \vee \neg q)) \vee (p \wedge q)]$	Distributive Law
3	$\neg[(\neg p \wedge T) \vee (p \wedge q)]$	Complement Law: $q \vee \neg q = T$
4	$\neg[\neg p \vee (p \wedge q)]$	Identity Law: $(\neg p \wedge T) = \neg p$
5	$\neg[(\neg p \vee p) \wedge (\neg p \vee q)]$	Distributive Law
6	$\neg[T \wedge (\neg p \vee q)]$	Complement Law: $(\neg p \vee p) = T$
7	$\neg(\neg p \vee q)$	Identity Law: $T \wedge X = X$
8	$p \wedge \neg q$	De Morgan's Law
Final Simplified Expression	$p \wedge \neg q$	Final Answer

Part c)

At a university sports club, the championship trophy was stolen by one of four students: **Ayan**, **Bilal**, **Daniyal**, or **Hamza**.

Professor Reed investigated the case and discovered that:

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Only one student was telling the truth, while the other three were lying.

The students made the following statements:

Student	Statement
Ayan	“Bilal stole the trophy.”
Bilal	“Hamza did not steal the trophy.”
Daniyal	“Hamza was in the cafeteria with Ayan when the trophy disappeared.”
Hamza	“Bilal did not steal the trophy.”

Using logical reasoning, determine:

1. Which student stole the trophy.
2. Which student was telling the truth.

Step	Assumption / Analysis	Result
1	Only one student is telling the truth, and the other three are lying .	Important condition
2	Ayan says: “ Bilal stole the trophy. ”	If true → Bilal is thief
3	Hamza says: “ Bilal did not steal the trophy. ”	Opposite of Ayan’s statement
4	Since Ayan and Hamza give opposite statements, one of them must be telling the truth.	Exactly one of Ayan or Hamza is truthful
5	Assume Ayan is telling the truth . Then Bilal stole the trophy.	Check consistency
6	If Bilal stole the trophy, then Bilal’s statement “Hamza did not steal the trophy” is also true.	This gives two truthful students (Ayan and Bilal)
7	But only one student can tell the truth. Therefore, Ayan cannot be truthful.	Ayan is lying
8	Therefore, Hamza’s statement must be true: “ Bilal did not steal the trophy. ”	Hamza is truthful
9	Since Bilal is lying, his statement “Hamza did not steal the trophy” is false.	Therefore, Hamza stole the trophy
10	Daniyal says: “Hamza was in the cafeteria with Ayan when the trophy disappeared.” Since Daniyal is lying, this statement is false.	Consistent with conditions



Question no 3: Let the domains of discourse be: (10 marks)

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- x : all people
- y : all languages

Define the predicates:

- $S(x)$: x is a scholar
 - $A(y)$: y is an ancient language
 - $K(x,y)$: x knows language y
 - $R(y)$: language y is used to write a riddle in the secret book
 - $D(x)$: x can decode the secret book
1. Every scholar knows at least one ancient language.
 2. Every ancient language is used to write at least one riddle in the secret book.
 3. Anyone who knows an ancient language used in the secret book can decode the book.

Therefore, every scholar can decode the secret book.

1. Translate each premise and the conclusion into quantified symbolic form. (5 Marks)
2. Determine whether the argument is valid or invalid. Justify your answer using logical reasoning and rules of inference.(5 Marks)

Solution :

Symbolic Form of Premises and Conclusion

Statement	Symbolic Form
Every scholar knows at least one ancient language.	$\forall x (S(x) \rightarrow \exists y (A(y) \wedge K(x, y)))$
Every ancient language is used to write at least one riddle in the secret book.	$\forall y (A(y) \rightarrow R(y))$
Anyone who knows an ancient language used in the secret book can decode the book.	$\forall x \forall y ((K(x, y) \wedge R(y)) \rightarrow D(x))$
Conclusion: Every scholar can decode the secret book.	$\forall x (S(x) \rightarrow D(x))$

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Step	Logical Reasoning	Rule Used
1	Assume an arbitrary person x is a scholar: $S(x)$.	Assumption
2	From Premise 1: every scholar knows at least one ancient language. Therefore, there exists some language y such that: $A(y) \wedge K(x, y)$	Universal Instantiation
3	From Step 2, we obtain: $A(y)$ and $K(x, y)$	Simplification
4	From Premise 2 and $A(y)$: $R(y)$	Modus Ponens
5	Now we have: $K(x, y) \wedge A(y) \wedge R(y)$	Conjunction
6	Using Premise 3: $D(x)$	Modus Ponens
7	Since assuming $S(x)$ led to $D(x)$, we conclude: $S(x) \rightarrow D(x)$	Conditional Proof
8	Because x was arbitrary: $\forall x(S(x) \rightarrow D(x))$	Universal Generalization



Read each quantified statement carefully. Write its meaning in simple English and determine whether the statement is True or False according to the given scenario.

Music Streaming / Playlist Service Scenario

Users: Ali, Sana

Songs: Afreen Afreen, Jal Pari, Aitebar

Genres: Pop, Sufi/Folk, Electronic

Playlists: Study Time, Gym Mix

Facts:

1. Ali added Afreen Afreen to the Study Time playlist and listened to it. The song belongs to Pop.
2. Ali added Jal Pari to the Study Time playlist and listened to it. The song belongs to Sufi/Folk.
3. Sana added Aitebar to the Gym Mix playlist and listened to it. The song belongs to

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Electronic.

4. Sana did not add any song to the Study Time playlist.

5. Ali has never listened to any Electronic song.

6. Every song has exactly one genre.

Predicates:

$A(u,s,p)$: User u added song s to playlist p

$L(u,s)$: User u listened to song s

$G(s,g)$: Song s belongs to genre g

Student Activity Table

#	Quantified Statement	Simple English Meaning	True / False
1	$\forall s (A(\text{Ali},s,\text{StudyTime}) \rightarrow L(\text{Ali},s))$	Every song Ali added to Study Time was listened to by Ali.	True
2	$\forall s (G(s,\text{Electronic}) \rightarrow \neg L(\text{Ali},s))$	Ali never listened to any Electronic song.	True
3	$\forall s \neg A(\text{Sana},s,\text{StudyTime})$	Sana did not add any song to Study Time.	True
4	$\forall u \exists s A(u,s,\text{StudyTime})$	Every user added a song to Study Time.	False
5	$\exists s (G(s,\text{Electronic}) \wedge L(\text{Ali},s))$	There exists an Electronic song listened to by Ali.	false

[20 marks]

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Question 4) Part A)

Solution:

Suppose a, b , and c are integers and x, y , and z are nonzero real numbers that satisfy the following equations:

$$\frac{xy}{x+y} = a, \frac{xz}{x+z} = b, \frac{yz}{y+z} = c$$

Is x rational? If so, express it as ratio of two integers.

Proof. Taking the reciprocals of both sides of the first equation:

$$\frac{x+y}{xy} = \frac{1}{a}$$

Now we split the first fraction into two, and simplify:

$$\frac{x+y}{xy} = \frac{x}{xy} + \frac{y}{xy} = \frac{1}{y} + \frac{1}{x}$$

Therefore $\frac{1}{y} + \frac{1}{x} = \frac{1}{a}$. By performing the same steps on the other two equations, we see that $\frac{1}{z} + \frac{1}{x} = \frac{1}{b}$ and $\frac{1}{z} + \frac{1}{y} = \frac{1}{c}$.

For the sake of simplicity let's do some renaming: let

$$X = 1/x, Y = 1/y, Z = 1/z, A = 1/a, B = 1/b, C = 1/c$$

So the three new equations we derived above become:

$$Y + X = A \quad (1)$$

$$Z + X = B \quad (2)$$

$$Z + Y = C \quad (3)$$

From (1) we get (4): $Y = A - X$ and from (2) we get (5): $Z = B - X$.

Substituting (4) and (5) back into (3) we get: $(B - X) + (A - X) = C$.

So $B + A - 2X = C$, then $B + A - C = 2X$ and $\frac{1}{2}(B + A - C) = X$. Using our old variable names, we get:

$$\frac{1}{2} \left(\frac{1}{b} + \frac{1}{a} - \frac{1}{c} \right) = \frac{1}{x}$$

Rewriting:

$$\frac{1}{2b} + \frac{1}{2a} - \frac{1}{2c} = \frac{1}{x}$$

Getting a common denominator, then adding:

$$\frac{ac}{2abc} + \frac{bc}{2abc} - \frac{ab}{2abc} = \frac{ac + bc - ab}{2abc} = \frac{1}{x}$$

Finally, taking reciprocals of both sides, we get x as a ratio of two integers:

$$\frac{2abc}{ac + bc - ab} = x$$

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Part B) Investigate the divisibility constraints on integers and prove that the property of being divisible by 2 is an indispensable requirement for divisibility by 6.

Solution:

A necessary condition for an integer to be divisible by 6 is that it be divisible by 2.

Proof. Rewriting the statement, we get:

$\forall$ integers n , if n is divisible by 6 then n is divisible by 2.

True. Assume n is any integer divisible by 6. By definition of divisibility, $n = 6m$ for some integer m . Then $n = 6m = 2(3m)$. Let $t = 3m$. Then t is an integer because it is a product of integers. So $n = 2t$ where t is an integer, therefore n is divisible by 2 by the definition of divisibility. $\square$

Part C) Prove that the fourth power of any integer has the form $8m$ or $8m + 1$ for some integer m .

Solution:

Let n be any integer.

Case 1: n is even

$$n = 2k$$

Then:

$$n^4 = (2k)^4 = 16k^4 = 8(2k^4)$$

Hence n^4 has the form $8m$.

Case 2: n is odd

$$n = 2k + 1$$

Then:

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1$$

Since $k(k+1)$ is even:

$$4k^2 + 4k = 8t \text{ for some integer } t.$$

Thus:

$$n^2 = 8t + 1$$

Squaring again:

$$\begin{aligned} n^4 &= (8t+1)^2 \\ &= 64t^2 + 16t + 1 \\ &= 8(8t^2 + 2t) + 1 \end{aligned}$$

Hence n^4 has the form $8m + 1$.

Therefore:

$$n^4 = 8m \text{ or } 8m+1$$

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Part D) Prove that for all prime numbers a, b and c , $a^2 + b^2 \neq c^2$.

Solution:

For all prime numbers a, b , and c , $a^2 + b^2 \neq c^2$.

Hint: Use the fact that $a^2 = c^2 - b^2 = (c - b)(c + b)$ and apply the unique factorization of integers theorem.

Proof. Argue by contradiction and assume a, b, c are prime numbers such that $a^2 + b^2 = c^2$. Then $a^2 = c^2 - b^2 = (c - b)(c + b)$.

a^2 is factored in two ways, as $a \cdot a$ and as $(c - b)(c + b)$. So, by the unique factorization theorem, $c - b = a$ and $c + b = a$.

Adding these two equations together, we get $2c = 2a$ therefore $c = a$. But then $a^2 + b^2 = a^2$ by substitution, so $b = 0$, which is not a prime number, a contradiction.

[Therefore our supposition was false, and the original statement is true.] □

Part E) In a group of 15 people, is it possible for each person to have exactly 3 friends? Justify your answer. (Assume that friendship is a symmetric relationship: If x is a friend of y , then y is a friend of x .)

Solution:

Represent people as vertices of a graph.

Each person has exactly 3 friends.

Total degree sum:

$$15 \times 3 = 45$$

By the Handshaking Theorem, the sum of degrees must be even.

But 45 is odd.

Contradiction.

Therefore such a friendship arrangement is impossible.

[CLO3] Apply mathematical induction to prove properties of sequences, recursive relations.

Question no 5: (4+4+4+4) (16 Marks)

Part A) A chain letter works as follows: One person sends a copy of the letter to five friends, each of whom sends a copy to five friends, each of whom sends a copy to five friends, and so forth. How many people will have received copies of the letter after the twentieth repetition of this process, assuming no person receives more than one copy?

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A chain letter works as follows: One person sends a copy of the letter to five friends, each of whom sends a copy to five friends, each of whom sends a copy to five friends, and so forth. How many people will have received copies of the letter after the twentieth repetition of this process, assuming no person receives more than one copy?

Proof. $\sum_{k=0}^{20} 5^k = \frac{5^{21} - 1}{4} \approx 1.192 \times 10^{14} \approx 119 \text{ trillion people}$ (this is about 20,000 times the current population of the Earth!) $\square$

Part B)

Suppose that $e_0, e_1, e_2, \dots$ is a sequence defined as follows:

$$e_0 = 12, \quad e_1 = 29,$$

$$e_k = 5e_{k-1} - 6e_{k-2} \text{ for each integer } k \geq 2.$$

Prove that $e_n = 5 \cdot 3^n + 7 \cdot 2^n$ for every integer $n \geq 0$.

Solution:

Suppose that $e_0, e_1, e_2, \dots$ is a sequence defined as follows:

$$e_0 = 12, e_1 = 29, e_k = 5e_{k-1} - 6e_{k-2} \text{ for every integer } k \geq 2.$$

Prove that $e_n = 5 \cdot 3^n + 7 \cdot 2^n$ for every integer $n \geq 0$.

Proof. Let the property $P(n)$ be the equation $e_n = 5 \cdot 3^n + 7 \cdot 2^n$.

Show that $P(0)$ and $P(1)$ are true:

We must show that $e_0 = 5 \cdot 3^0 + 7 \cdot 2^0$ and $e_1 = 5 \cdot 3^1 + 7 \cdot 2^1$. The left-hand side of the first equation is 12 (by definition of $e_0, e_1, e_2, \dots$), and its right-hand side is $5 \cdot 1 + 7 \cdot 1 = 12$ also. The left-hand side of the second equation is 29 (by definition of $e_0, e_1, e_2, \dots$), and its right-hand side is $5 \cdot 3 + 7 \cdot 2 = 29$ also. Thus $P(0)$ and $P(1)$ are true.

Show that for every integer $k \geq 1$, if $P(i)$ is true for each integer i with $0 \leq i \leq k$, then $P(k+1)$ is also true:

Let k be any integer with $k \geq 1$ and suppose that $e_i = 5 \cdot 3^i + 7 \cdot 2^i$ for each integer i with $0 \leq i \leq k$. [This is the inductive hypothesis.] We must show that $e_{k+1} = 5 \cdot 3^{k+1} + 7 \cdot 2^{k+1}$. Now

$e_{k+1} = 5e_k - 6e_{k-1}$	by definition of $e_0, e_1, e_2, \dots$
$= 5(5 \cdot 3^k + 7 \cdot 2^k) - 6(5 \cdot 3^{k-1} + 7 \cdot 2^{k-1})$	by inductive hypothesis
$= 25 \cdot 3^k + 35 \cdot 2^k - 30 \cdot 3^{k-1} - 42 \cdot 2^{k-1}$	by distributing
$= 25 \cdot 3^k + 35 \cdot 2^k - 10 \cdot 3 \cdot 3^{k-1} - 21 \cdot 2 \cdot 2^{k-1}$	by algebra
$= 25 \cdot 3^k + 35 \cdot 2^k - 10 \cdot 3^k - 21 \cdot 2^k$	by laws of exponents
$= 15 \cdot 3^k + 14 \cdot 2^k$	by combining like terms
$= 5 \cdot 3 \cdot 3^k + 7 \cdot 2 \cdot 2^k$	by algebra
$= 5 \cdot 3^{k+1} + 7 \cdot 2^{k+1}$	by laws of exponents

[as was to be shown]. $\square$

Part C)

[Pre – condition: $\text{largest} = A[1]$ and $i = 1$]

while ($i \neq m$)

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```
1.  $i := i + 1$   
2. if  $A[i] > largest$  then  $largest := A[i]$ 
```

end while

[Post – condition: $largest = \text{maximum value of } A[1], A[2], \dots, A[m]$]

loop invariant: $I(n)$ is “ $largest = \text{maximum value of } A[1], A[2], \dots, A[n + 1]$ and $i = n + 1.$ ”

Use the loop invariant theorem to prove the correctness of the loop with respect to the pre- and post-conditions.

Solution:

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[Pre-condition: $\text{largest} = A[1]$ and $i = 1$].

```
while ( $i \neq m$ )  
    1.  $i := i + 1$   
    2. if  $A[i] > \text{largest}$  then  $\text{largest} := A[i]$   
end while
```

[Post-condition: $\text{largest} = \text{maximum value of } A[1], A[2], \dots, A[m]$.]

loop invariant: $I(n)$ is “ $\text{largest} = \text{maximum value of } A[1], \dots, A[n+1]$ and $i = n + 1$.”

Proof. I. Basis Property: [$I(0)$ is true before the first iteration of the loop.]

$I(0)$ is “ $\text{largest} = \text{maximum of } A[1]$ and $i = 0 + 1$.” According to the pre-condition, before the first iteration of the loop, $\text{largest} = A[1]$ and $i = 1$. So $I(0)$ is evidently true.

II. Inductive Property: [If $G \wedge I(k)$ is true before a loop iteration (where $k \geq 0$), then $I(k + 1)$ is true after the loop iteration.]

Suppose k is any nonnegative integer such that $G \wedge I(k)$ is true before an iteration of the loop. Then as execution reaches the top of the loop, $i \neq m$, $i = k + 1$ and $\text{largest} = \text{maximum of } A[1], \dots, A[k + 1]$. Since $i \neq m$, the guard is passed and statement 1 is executed. The execution of statement 1 has the following effect:

$$i_{\text{new}} = i_{\text{old}} + 1 = k + 1 + 1 = k + 2.$$

Similarly, before statement 2 is executed, $\text{largest}_{\text{old}} = \text{maximum of } A[1], \dots, A[k + 1]$, so after execution of statement 2, if $A[k + 2] > \text{largest}$ then we have

$$\text{largest}_{\text{new}} = A[k + 2]$$

and else we have

$$\text{largest}_{\text{new}} = \text{largest}_{\text{old}}.$$

Hence after the loop iteration, the two statements $\text{largest} = \text{maximum of } A[1], \dots, A[k + 2]$ and $i = (k + 1) + 1$ are true, and so $I(k + 1)$ is true.

III. Eventual Falsity of Guard: [After a finite number of iterations of the loop, G becomes false.]

The guard G is the condition $i \neq m$, and m is a nonnegative integer. By I and II, it is known that for every integer $n \geq 0$, if the loop is iterated n times, then $i = n + 1$. So after $m - 1$ iterations of the loop, $i = m$. Thus G becomes false after $m - 1$ iterations of the loop.

IV. Correctness of the Post-Condition: [If N is the least number of iterations after which G is false and $I(N)$ is true, then the value of the algorithm variables will be as specified in the post-condition of the loop.]

According to the post-condition, the value of largest after execution of the loop should be the maximum of $A[1], \dots, A[m]$. But when G is false, $i = m$. And when $I(N)$ is

true, $i = N + 1$ and $\text{largest} = \text{maximum of } A[1], \dots, A[N + 1]$. Since both conditions (G false and $I(N)$ true) are satisfied, $m = i = N + 1$ and so $\text{largest} = \text{maximum of } A[1], \dots, A[m]$, as required. $\square$

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Part D)

Let $t_0, t_1, t_2, \dots$ be defined by the formula $t_n = 2 + n$ for every integer $n \geq 0$. Show that this sequence satisfies the following recurrence relation for every integer $k \geq 2$:

$$t_k = 2t_{k-1} - t_{k-2}$$

Solution:

Let $t_0, t_1, t_2, \dots$ be defined by the formula $t_n = 2 + n$ for every integer $n \geq 0$. Show that this sequence satisfies the following recurrence relation for every integer $k \geq 2$:
 $t_k = 2t_{k-1} - t_{k-2}$.

Proof. By definition of $t_0, t_1, t_2, \dots, t_n = 2 + n$ for each integer $n \geq 0$. Substitute $k, k-1$ and $k-2$ in place of n to get

$$\begin{aligned} (*) \quad t_k &= 2 + k, \\ (**) \quad t_{k-1} &= 2 + (k-1), \\ (***) \quad t_{k-2} &= 2 + (k-2) \end{aligned}$$

for each integer $k \geq 2$. Then $2t_{k-1} - t_{k-2}$

$$\begin{aligned} &= 2(2 + k - 1) - (2 + k - 2) && \text{by substitution from (**) and (***)} \\ &= 2(k + 1) - k && \text{by basic algebra} \\ &= k + 2 && \text{by basic algebra} \\ &= t_k && \text{by substitution from (*)} \end{aligned}$$

□

Question no 6)

Solution

Given the city-count threshold:

K = 15

Two items are connected if the number of cities with similar price patterns is greater than or equal to 15.

—

1: Construct the Graph

Given data:

Item Pair	Number of Cities
Flour–Rice	18
Flour–Sugar	5
Rice–Sugar	14
Milk–Tea	20

We compare each value with K = 15.

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Pair Analysis

- * Flour–Rice = 18
Since $18 \geq 15$, an edge exists.
- * Flour–Sugar = 5
Since $5 < 15$, no edge exists.
- * Rice–Sugar = 14
Since $14 < 15$, no edge exists.
- * Milk–Tea = 20
Since $20 \geq 15$, an edge exists.

—

Resulting Graph

Edges:

$E = \{(Flour, Rice), (Milk, Tea)\}$

Graph structure:

Flour ----- Rice
Milk ----- Tea
Sugar

Sugar is an isolated node.

—

2: Degree of Each Node

The degree of a node is the number of edges connected to it.

Item	Degree
Flour	1
Rice	1
Sugar	0
Milk	1
Tea	1

—

3: Most Central Item Based on Degree

Degree centrality is based on the number of direct connections.

Here:

- * Flour, Rice, Milk, and Tea each have degree 1.

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* Sugar has degree 0.

Therefore, the most central items are:

$\boxed{\text{Flour, Rice, Milk, and Tea}}$

because they all have the highest degree.

—

4: Effect of Reducing K to 10

K = 10

Re-evaluate the pairs.

Item Pair	Cities	Edge?
Flour–Rice	18	Yes
Flour–Sugar	5	No
Rice–Sugar	14	Yes
Milk–Tea	20	Yes

The new edge added is:

(Rice, Sugar)

—

New Graph

Flour ----- Rice ----- Sugar
Milk ----- Tea

—

Updated Degrees

Item	Degree
Flour	1
Rice	2
Sugar	1
Milk	1
Tea	1